



Transport of coexisting Ni-Cu sulfide liquid and silicate melt in partially molten peridotite

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ARTICLE INFO

Article history:

Received 10 October 2019

Received in revised form 6 February 2020

Accepted 13 February 2020

Available online 20 February 2020

Editor: J. Brodholt

Keywords:

liquid extraction

lithosphere refertilization

magmatic sulfide deposit

high P-T experiment

synchrotron X-ray microtomography

ABSTRACT

Transport of coexisting sulfide and silicate melts in partially molten peridotite contributes to the redistribution of chalcophile elements within the upper mantle as well as the genesis of magmatic Ni-Cu sulfide deposits, but has not been investigated systematically. Using laboratory experiments, theoretical calculations, and X-ray synchrotron microtomography, this study documents the topology and considers controls on the extraction of two immiscible liquids during partial melting of mantle peridotite. Under hydrostatic conditions, the measured dihedral angle at silicate melt-mineral-mineral junctions is 13.7–21.3°. Silicate melt is distributed along grain edges forming incompletely interconnected melt channels that disconnected by some dead ends at melt fractions ~7–9 vol%. Application of theoretically predicted permeability ($k \sim 10^{-14}$ – 10^{-16} m²) permits estimation of the extraction velocity of silicate melt of 0.7–11.1 μm/day within a single interconnected melt channel. In the absence of silicate melt, isolated sulfide droplets (3.77 vol%) show a sulfide-olivine-olivine dihedral angle of 91.5–101.3°. However, in the presence of silicate melt, sulfide droplets (average size $\sim 2.53 \pm 2.14$ μm, 1σ) are partially surrounded by silicate melt and stranded in triple junctions or melt pockets due to the limitation of the smallest dimension (0.3 μm) of melt channels. Thus, the extraction of sulfide liquid is highly restricted by these dead ends and the smallest dimension of melt channels during porous flow of silicate melt. In contrast, during large-strain shear deformation (strain ~1.6–2.5), initially stranded sulfide droplets were elongated and extracted with silicate melt into liquid-rich sheets with a length of several hundred microns, constantly oriented at $14.3 \pm 4.5^\circ$ to the shear plane and antithetic to the shear direction. The angle is lower than that (18–30°) of those sheets containing sulfide liquid only, indicating that silicate melt dominates liquid-rich sheets. Driven by stress, silicate melt-dominated liquid-rich sheets open the appropriately oriented grain boundaries between silicate minerals, thus providing an efficient pathway for the extraction of sulfide liquid during deformation. When such sheet-like channels remain open, sulfide droplets (> millimeter-scale) can be potentially mobile through high strain domains of the upper mantle, contributing to the addition of chalcophile elements and the fertilization of the lithospheric mantle.

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1. Introduction

It has been well documented that the formation of magmatic Ni-Cu sulfide deposits responds to the saturation of sulfur in mantle-derived basic and ultrabasic magmas followed by

the segregation and concentration of immiscible sulfide liquids (Arndt et al., 2005). Notable examples of such deposits occur under craton margin conditions, which are closely associated with the thin lithospheric mantle (Griffin et al., 2013). Moreover, it has been proposed that the depleted lithospheric mantle can undergo refertilization or modification during the transit of multi-stage asthenosphere-derived melts containing ore-forming elements (Griffin et al., 2004). The refertilization of the depleted mantle by the later addition of sulfide liquids has been also observed in mantle peridotites from xenoliths or orogenic belts (Lorand et al., 2010). However, this process remains controversial and poorly documented (e.g., Riches and Rogers, 2011), and it remains un-

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clear whether or how sulfide liquid can be physically transported through the lithospheric mantle. To the extent that “re-fertilized” mantle domains might contribute to the genesis of magmatic ore deposits, it is important to understand physical controls on the sulfide-liquid migration through peridotites at low silicate-melt fractions.

In partially molten rocks, the dihedral angle (θ) of the solid matrix-liquid system is widely assumed to be a key factor in influencing the migration behavior of liquids because of dihedral angles at liquid-solid-solid triple junctions fundamentally controlling the pore morphology (von Bargen and Waff, 1986). The dihedral angle θ is defined as:

$$2\cos\frac{\theta}{2} = \frac{\gamma_{s-s}}{\gamma_{s-l}}, \quad (1)$$

where γ_{s-s} and γ_{s-l} are respectively the solid-solid and solid-liquid interfacial energy per unit area. A dihedral angle (θ) below 60° leads to a regular interconnected liquid network along grain edges in an ideal system (Zhu et al., 2011), even if the liquid fraction is very low (0.5 vol%, Ghanbarzadeh et al., 2017). Conversely, at $\theta > 60^\circ$, liquid is isolated and trapped at grain corners when the liquid fraction is lower than percolation thresholds (e.g., for Fe-S or Fe-S-O liquid, 17.5 vol%, Bagdassarov et al., 2009a; 6–9.5 vol%, Terasaki et al., 2005; 3–6 vol%, Yoshino et al., 2003), but when percolation thresholds are exceeded, a liquid network is formed in a solid matrix and may remain connected down to a very low liquid fraction (~ 1 –2%) during drainage due to the hysteretic percolation of liquid with $\theta > 60^\circ$ (Ghanbarzadeh et al., 2017). The dihedral angle and consequent percolation threshold are known to vary as functions of intensive parameters such as the liquid composition, oxygen and sulfur fugacities (Gaetani and Grove, 1999; Holzheid et al., 2000). During shear deformation of a partially molten peridotite, the distribution of silicate melt is highly heterogeneous, forming stress-driven melt-rich bands or sheets (Holtzman and Kohlstedt, 2007; Soustelle et al., 2014) oriented at a low angle (15 – 20°) to the shear plane, which is consistent with the numerical model analysis of viscous anisotropy (Takei and Katz, 2015).

The distribution and extraction of sulfide liquids with large interfacial energy and complex chemical compositions are determined by the interplay of surface tension and stress (Walte et al., 2011) and control the enrichment of ore metals (Cu, Ni, PGEs, etc) due to very high sulfide-silicate melt partition coefficients of these elements (e.g., Mungall and Brenan, 2014). Some hydrostatic experiments have shown that sulfide liquids with high sulfide-olivine dihedral angles ($>60^\circ$) are commonly trapped within a solid silicate-matrix at upper mantle pressure-temperature conditions as long as the liquid fraction does not exceed percolation thresholds (Holzheid, 2013; Holzheid et al., 2000). In contrast, driven by shear stress, initially stranded sulfide droplets are elongated to form sulfide-rich bands (Hustoft and Kohlstedt, 2006) or alignments of elongated sulfide droplets (Bruhn et al., 2000), which are high-permeability pathways for the extraction of sulfide liquid from a solid silicate-matrix. Similarly, stress can also assist the percolation of hydrocarbon-bearing fluids ($\theta > 60^\circ$) in the statically impermeable rock salt (Ghanbarzadeh et al., 2015).

However, upon overstepping the silicate solidus, the extraction of sulfide liquid can be further complicated because of the interaction between silicate melt and sulfide liquid. Under hydrostatic conditions, most sulfide droplets cannot be entrained through the porous flow of silicate melt due to the larger excess pressure imposed by the capillary force, low flow rates of silicate melt and a higher percolation threshold (13 ± 2 vol%) in the presence of low to moderate degrees of silicate melt (Holzheid et al., 2000; Mungall and Su, 2005; Yoshino et al., 2004). Under low-strain axial compression conditions, several studies (Cerantola et al., 2015; Walte

et al., 2011) have suggested that silicate melt may impede the mechanical extraction and segregation of sulfide liquid. In contrast, Bagdassarov et al. (2009b) argued that a higher degree of partial silicate melting enhances the segregation velocity of Fe-S liquid in a peridotite based on the theoretical calculation of segregation velocities and the distribution of the Fe-S and silicate liquids in centrifuge experiments. Therefore, to systematically investigate the extraction of sulfide liquid after the presence of silicate melt and provide the experimental evidence for the fertilization of lithospheric mantle and the genesis of magmatic sulfide deposits, we carried out hydrostatic and shear deformation experiments and characterized the three-dimensional distribution of two immiscible liquids in experimental samples using postmortem X-ray synchrotron microtomography and theoretical calculations.

2. Materials and methods

2.1. Starting materials

Three types of starting materials were selected: (1) olivine + Ni-Cu sulfides, (2) pyrolite +/- Ni-Cu sulfides, and (3) olivine + basalt + Ni-Cu sulfides. Natural olivine crystals were separated from fresh spinel lherzolite xenoliths collected from Damaping (Hannuoba region), North China and were ground to 10 – $20 \mu\text{m}$ grain size in an agate mortar. Pyrolite from Jin et al. (1994) was ground to 5 – $15 \mu\text{m}$ grain size and used as the starting material of sulfide-free static experiments. Tholeiitic basalt from the East Pacific Rise (102.7044°W , 2.64961°S) was ground to $<10 \mu\text{m}$ grain size. Ni-Cu sulfides were composed of pyrrhotite (59 wt%), pentlandite (36 wt%), and chalcopyrite (5 wt%) separates, similar to the composition of base-metal sulfide aggregates in massif peridotites (Lorand et al., 2010), and were ground to $<10 \mu\text{m}$. For static experiments containing solids and two immiscible liquids, 1 wt% of Ni-Cu sulfides were mixed with 99 wt% of the pyrolite; for static experiments containing only solids and sulfide liquid, a mixture of 5 wt% Ni-Cu sulfides and 95 wt% olivine was used as the starting material. The starting material for deformation experiments was prepared using a mixture of 85 wt% olivine, 10 wt% basalt, and 5 wt% Ni-Cu sulfides. Compositions and densities of these starting materials are listed in Table 1.

2.2. Experimental methods

Static and deformation experiments were conducted at the State Key Laboratory of Geological Processes and Mineral Resources (GPMR) of China University of Geosciences using a 150-ton non-end-loaded type piston-cylinder press and a 5 GPa modified Griggs-type apparatus, respectively. For static experiments, starting materials were loaded into a 3.75 mm-diameter and 5–6 mm-high cylindrical platinum (Pt) capsule with a graphite inner sleeve (Fig. S3-right). A low-friction assembly consisting of NaCl and Pyrex sleeves, a graphite heater, sintered MgO spacers, and an Al_2O_3 plug, was used for the piston-cylinder experiments. Pressure calibration was conducted against the quartz/coesite phase transition and is accurate to ± 0.1 GPa. The temperature was monitored by a W/Re type C thermocouple located at the bottom of the capsule. All experiments were conducted under a pressure of 1.5 GPa and temperatures of 600 to 1300°C for 12 h prior to being quenched to room temperature by shutting off power to the furnace. For shear experiments, thin (400 – $650 \mu\text{m}$ -thick) elliptical slices of the hot-pressed olivine + basalt + sulfide mixture were prepared at 1.5 GPa, 1100°C below the expected solidus of the Fe-Ni-Cu sulfide mixture (1.5 GPa, 1150°C , Zhang and Hirschmann, 2016) and were placed between the halves of a 9 mm-long Al_2O_3 piston cut at an angle of 45° in the middle (Fig. S3-left). Shear experiments were performed at 1.5 GPa, 1250°C and a strain rate of 10^{-4} s^{-1} . CsCl

Table 1

The chemical composition and density of starting materials.

wt%	Pyrolite (Jin et al., 1994)				Basalt	wt%	Ni-Cu sulfides		
	Ol (52 wt%)	Opx (31 wt%)	Cpx (14 wt%)	Sp (3 wt%)			Ccp (5 wt%)	Po (59 wt%)	Pn (36 wt%)
SiO ₂	41.42(31)	56.75(13)	52.66(20)	0.03(1)	51.73(24)	S	35.29(14)	38.63(56)	34.06(41)
TiO ₂	0.02(2)	0.08(2)	0.38(16)	0.05(2)	2.30(6)	Fe	30.23(27)	59.58(36)	33.04(88)
Al ₂ O ₃	0.01(1)	3.91(23)	5.24(55)	56.65(23)	12.85(5)	Ni	0.10(13)	0.13(3)	32.18(65)
Cr ₂ O ₃	0.01(1)	0.32(3)	0.82(9)	11.37(10)	0 ^b	Cu	33.46(15)	0.01(1)	0 ^b
MgO	48.89(15)	32.55(21)	16.85(55)	19.42(19)	4.62(3)	$\rho_{\text{Pyrolite}} = 3050 \text{ kg/m}^3$ $\rho_{\text{olivine}} = 3200 \text{ kg/m}^3$ $\rho_{\text{basalt}} = 2600 \text{ kg/m}^3$ $\rho_{\text{Ni-Cu sulfides}} = 4300 \text{ kg/m}^3$ $\Delta\rho_s = 1100 \text{ kg/m}^3$			
CaO	0.04(2)	0.50(5)	20.35(48)	0.02(2)	8.33(1)				
MnO	0.15(2)	0.13(1)	0.11(2)	0.13(1)	0.23(1)				
FeO ^a	8.84(28)	5.84(14)	2.66(12)	10.56(12)	14.60(36)				
Na ₂ O	0.01(1)	0.06(1)	1.13(27)	0.01(1)	3.59(4)				
NiO	0.31(2)	0.12(03)	0.03(04)	0.43(1)	0 ^b				
K ₂ O	0.01(1)	0 ^b	0 ^b	0.01(1)	0.33(1)				
H ₂ O ⁺	- ^c	- ^c	- ^c	- ^c	1.19(26)				

The component of basalt was analyzed by XRF, and components of three sulfide minerals and four silicate minerals were analyzed by EPMA using an accelerating voltage of 15 kV, a beam size of 1.0–2.0 μm , and a beam current of 20 nA, respectively. 1σ error is given in parentheses. Mineral abbreviations: Ol-olivine, Opx-orthopyroxene, Cpx-clinopyroxene, Sp-spinel, Ccp-chalcocopyrite, Po-pyrrhotite, Pn-pentlandite. ^aTotal FeO; ^bLower the detection limit; ^cNot analyzed. The density of Ni-Cu sulfides ($\rho_{\text{Ni-Cu sulfides}}$) is calculated based on the model from Kress et al. (2008). $\Delta\rho_s$ is the density contrast between Ni-Cu sulfides and olivine.

was used as the confining medium. Specimens were pressurized to 1.5 GPa and kept at target temperatures for 2 h before deformation. The temperature was monitored by two W/Re type C thermocouples, with one being near the top of the specimen and the other near the bottom. Before being sealed, all capsules were heated at 120 °C in a vacuum oven for more than 12 h to remove adsorbed water vapor in the specimens. No apparent sulfur loss is observed during the drying (Supplementary material).

2.3. Analytical techniques

Polished sections cut parallel to the specimen axis were prepared from the recovered experimental specimens. The microstructure of experimental run products was observed using a Quanta 450 field-emission scanning electron microscope (FE-SEM) at the GPMR with an accelerating voltage of 20 kV, a spot size of 6.0 and a working distance of 12 mm. Quantitative compositional analyses were performed using an electron probe microanalyzer (EPMA, JXA-8100) at the Key Laboratory of Submarine Geosciences, State Oceanic Administration, China and the X-ray fluorescence (XRF) spectrometry at the ALS Minerals-ALS Chemex (Guangzhou) Co Ltd, China.

The crystallographic preferred orientations (CPOs) and shape preferred orientations (SPOs) of olivine were analyzed by indexing of electron backscattered diffraction (EBSD) patterns using a Quanta 450 FE-SEM equipped with an HKL Nordlys EBSD detector at the GPMR. An accelerating voltage of 20 kV, a spot size of 6, a beam current of 6 nA and a working distance of 20–25 mm were used for automatic mapping measurements.

Microtomography measurements were conducted at 2-BM of the Advanced Photon Source at Argonne National Laboratory, Argonne, IL, USA. The tomographic scans were done with monochromatic X-ray of 24.4 keV. In a scan, the sample was mounted on top of a rotary stage and rotated through a range of 180°. X-ray projection images of the sample were acquired every 0.12°, with a lens-coupled scintillator-based X-ray camera. The sample-to-detector distance was 100 mm for every scan. The projection images of each sample were used in tomographic structure reconstructions with Tomopy, a tomography data processing package (Supplementary material). The voxel size in the tomographic reconstructions is 0.87 μm isotropic in all three dimensions.

3. Results

3.1. Two-dimensional distribution of two immiscible liquids under hydrostatic conditions

Two sets of hydrostatic experiments were performed to compare the distribution of sulfide liquid under silicate melt-free and silicate melt-bearing conditions (Table 2). All two-dimensional images were analyzed using the freeware ImageJ developed by the National Institute of Health (NIH) (<http://imagej.nih.gov/ij/>), the analysis methodology is described in the supplementary material.

Under low-temperature (650 °C) and silicate melt-free conditions, sulfide mineral grains were generally polygonal and uniformly filled in pores among olivine grains. When temperatures were higher than the solidus (1150 °C, Zhang and Hirschmann, 2016) of sulfides, isolated sulfide droplets were observed within liquid-filled convex-outward triple junctions or pockets with a medium sulfide-olivine-olivine dihedral angle of 91.5–101.3° (95% confidence interval) (Fig. 1a and Fig. S4a) that is similar to results (median ~90–106°) of previous studies at similar pressures (1–2 GPa) and higher temperatures (1350–1500 °C) (Holzheid, 2013; Holzheid et al., 2000). The distribution of sulfide liquid was homogeneous, and the sulfide-liquid fraction was $3.70 \pm 0.16 \text{ area\%}$ (1σ) (the raw data are listed in supplementary files) based on analyses of two-dimensional SEM images, which is consistent with the theoretical estimation (3.77 vol%) based on compositions as weighted in Table 2. These straight grain boundaries and triple junctions of 120° among olivine grains (Fig. 1a) both indicated that the local equilibrium was obtained during an annealing time of 12 h (Jin et al., 1994).

Under silicate melt-bearing conditions, silicate melt was mainly distributed in triple junctions or melt pockets (Fig. 1b and c-white arrows). Moreover, at higher temperatures (1250–1300 °C), more silicate melt was present to form melt channels between two grains (Fig. 1c and d-white dotted lines), which may be two-grain boundaries or longitudinal sections of triple junctions. The sulfide liquid formed isolated and roundish droplets (Fig. 1b, c and d-red arrows) that were trapped in silicate-melt-filled triple junctions or pockets. Triple junctions and melt pockets commonly contained more than one sulfide droplet at 1300 °C (Fig. 1d), however we would anticipate that longer durations would lead to

Table 2

Summary of experimental conditions and liquid distributions.

No.	Starting materials	Experimental conditions	Liquid distributions
Hydrostatic experiments (12 h)			
GA122	Olivine + sulfide (3.77 vol% ^a)	650 °C	Sulfide liquid-free
GA159		1200 °C	Isolated sulfide droplets at triple junctions/pockets with $\theta \sim 91.5\text{--}101.3^\circ$.
GA115		1250 °C	
PC184	Pyrolite	1200 °C	Silicate melt forming locally connected tubular melt-channels along grain edges.
PC225		1250 °C	
PC235		1300 °C	
PC186	Pyrolite + sulfide (0.72 vol% ^a)	600 °C	Sulfide liquid-free
PC197		950 °C	
PC187		1050 °C	
PC196		1150 °C	Silicate melt with $\theta \sim 13.7\text{--}21.3^\circ$ forming locally connected tubular melt-channels along grain edges. Isolated sulfide droplets surrounded by silicate melt and trapped in the triple junctions/pockets.
PC185		1200 °C	
PC190		1250 °C	
PC226		1250 °C	
PC236		1300 °C	
Shear deformation experiments (1250 °C)			
GA383	Olivine + sulfide (3.68 vol% ^a) + basalt	$\dot{\epsilon} \sim 10^{-4} \text{ s}^{-1}$, $\gamma \sim 2.5$; $\sigma \sim 149 \pm 12 \text{ MPa}$	Silicate melt and sulfide pockets elongated and distributed alternately in liquid-rich sheets.
GA394		$\dot{\epsilon} \sim 10^{-4} \text{ s}^{-1}$, $\gamma \sim 1.6$; $\sigma \sim 147 \pm 8 \text{ MPa}$	

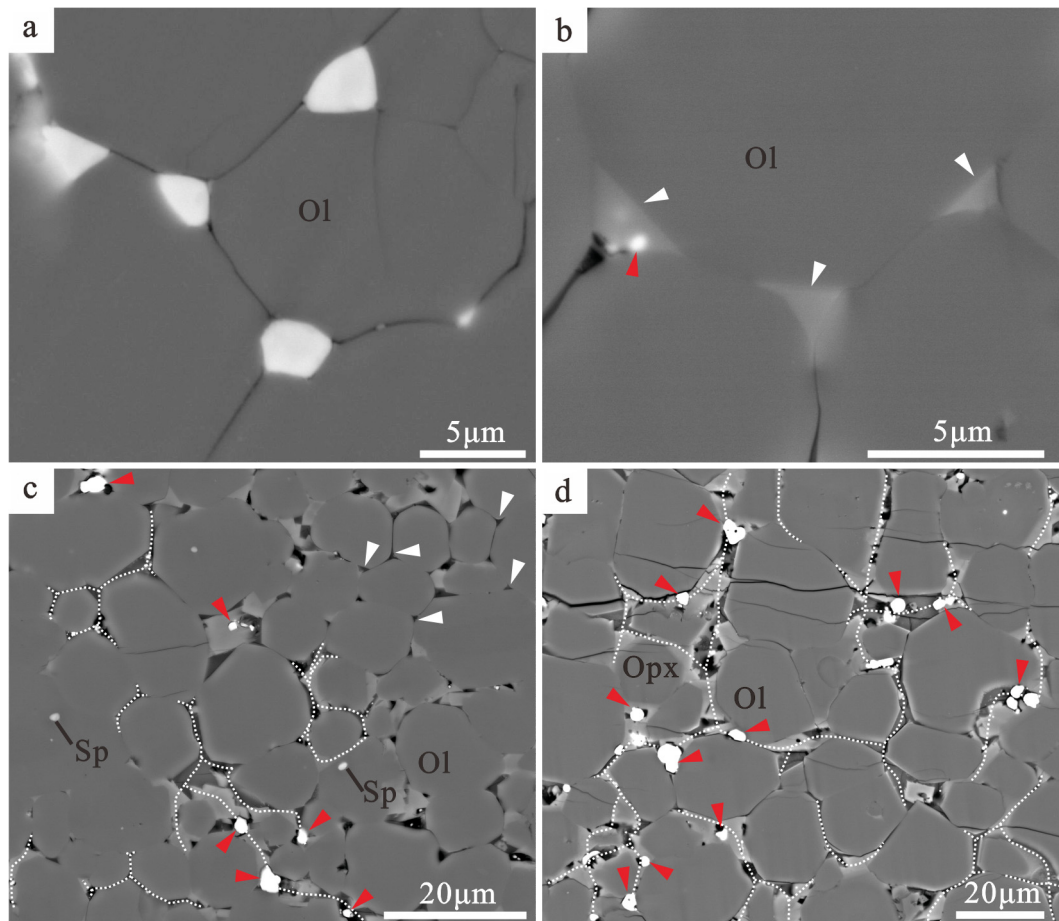
All experiments were conducted at a constant pressure of 1.5 GPa. σ -stress; γ -shear strain; $\dot{\epsilon}$ -strain rate; θ -apparent dihedral angle.^aCalculated volume fraction based on compositions as weighted (these values of the density of olivine, pyrolite and Ni-Cu sulfides are listed in Table 1).

Fig. 1. Back-scattered electron images of experimental products from hydrostatic experiments. (a) The distribution of sulfide liquid (white phase) in the experiment of olivine + sulfide system (GA115, 1250 °C); (b–d) The distributions of sulfide liquid and quenched silicate melt in experiments of pyrolite + sulfide system (b-PC185, 1200 °C; c-PC190, 1250 °C; d-PC236, 1300 °C). Red and white arrows respectively indicate sulfide droplets and silicate melt-filled triple junctions, and silicate melt channels are marked by white dotted lines. Mineral abbreviations: Ol-olivine; Opx-orthopyroxene; Sp-spinel. Note the possible melt film (thickness $\sim$ submicron-scale) between olivine grains in the upper right corner of Fig. 1c. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

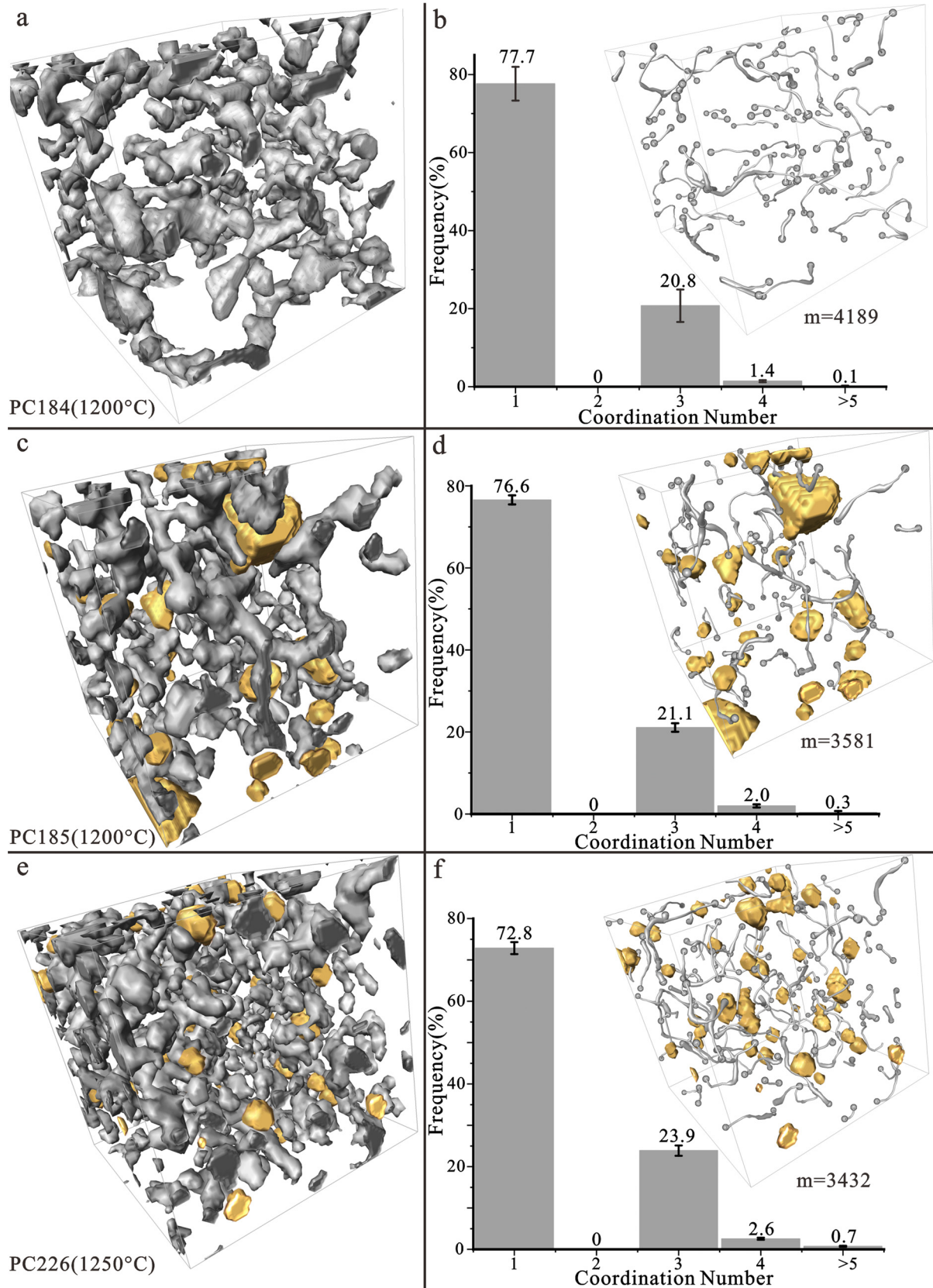


Fig. 2. Representative three-dimensional (3D) distributions of sulfide liquid and silicate melt and the corresponding statistical results of the node coordination number (the silicate melt connectivity) in experiments PC184 (a and b), PC185 (c and d) and PC226 (e and f). Insets in the right panel show corresponding skeletons of silicate melt channels. m is the number of statistical nodes. Gray channels and yellow spheres indicate silicate melt channels and sulfide droplets, respectively. Vacant volumes are occupied by silicate mineral grains. The size of each cube is 50 μm by 50 μm by 50 μm . See Video S1 in supplementary files for the movie of the 3D liquid distribution (the spatial resolution $\sim 0.87 \mu\text{m}$ in all three dimensions).

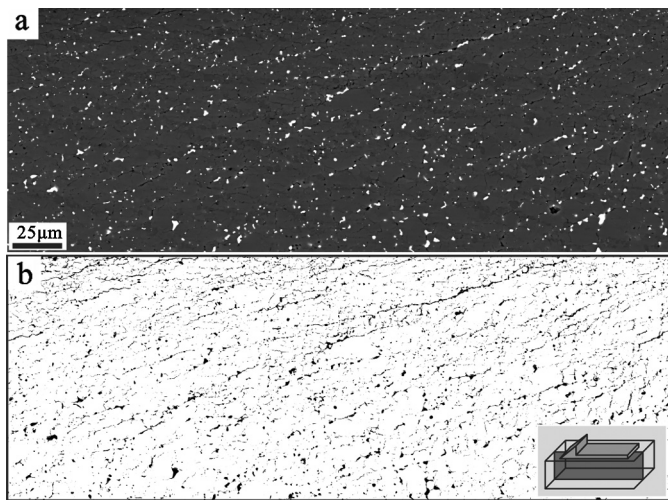


Fig. 3. Overview (a) of the distribution of two immiscible liquids (back-scattered electron image) and the binary image (b) of the same region to highlight the liquid distribution in experiment GA383 (strain rate $\sim 10^{-4}$ s $^{-1}$, strain ~ 2.5). Silicate melt and sulfide liquid in (a) are respectively indicated by the black and white phase, and the gray phase marks silicate minerals (olivine and orthopyroxene). The section of the sample is in a plane perpendicular to the shear plane and parallel to the shear direction, as illustrated by the inset in the lower right of the image (b). Two Al₂O₃ pistons (not shown) are respectively placed on the top and bottom of the sample forming low-permeability contact regions between the sample and pistons.

the consolidation of multiple sulfide droplets to leave at most one within each melt pocket (Mungall and Su, 2005). With increasing temperature, the dihedral angle (median) of silicate melt against minerals decreased from 19.3–21.3° (95% confidence interval) at 1200 °C to 13.7–15.5° (95% confidence interval) at 1250 °C (Fig. S4b and c; the raw data are listed in supplementary files), and the silicate-melt fraction increased from 9.01 ± 0.36 area% at 1200 °C to 12.00 ± 0.52 area% at 1300 °C, whereas the area fraction (1.17 ± 0.11 area%) and size (2.53 ± 2.14 μm) of sulfide droplets did not significantly vary with temperature within error (Fig. S4d; the raw data are listed in supplementary files). The experimental annealing time of 12 h may be sufficient for establishing a local equilibrium, as evidenced by smoothly curved solid-melt boundaries coexisting with flat solid-solid interfaces.

3.2. Three-dimensional distribution of two immiscible liquids under hydrostatic conditions

To understand the exact topology and transport of two immiscible liquids in mantle rocks, the three-dimensional data on the distribution of silicate melt and sulfide liquid in experiments PC184, PC185 and PC226 were obtained using X-ray synchrotron microtomography. The visualization of the three-dimensional topology of two immiscible liquids was conducted with the three-dimensional analysis software Avizo® (Fig. 2 and Video S1 in supplementary files; the analysis methodology is described in the supplementary material). The silicate melt network was dominated by tubular channels along grain edges in three-dimensional liquid distributions (Fig. 2a, c and e). Thus, most melt channels observed between two grains in two-dimensional SEM images should be interpreted as longitudinal sections of triple junctions. Sulfide liquid commonly formed irregular spheres or droplets that were partially surrounded by silicate melt and trapped by silicate-mineral grain faces in triple junctions and melt pockets (Fig. 2c and e). Based on these analyses for the topology of three-dimensional liquid distributions, the volume fraction of silicate melt was respectively 7.19 ± 0.24 vol% and 7.87 ± 0.19 vol% in experiment PC184 (sulfide-free, 1200 °C) and PC185 (sulfide-bearing, 1200 °C) and increased to 9.17 ± 0.13 vol% in experiment PC226 (sulfide-bearing,

1250 °C), and the volume fraction of sulfide liquid was respectively 0.83 ± 0.13 vol% and 0.68 ± 0.06 vol% in experiment PC185 and PC226, which are consistent with the theoretical estimation (0.72 vol%) (Table 2). Melt skeletons and nodes were obtained (Fig. 2b, d and f) using the Avizo®'s skeletonization module that systematically reduces cross-sections orthogonal to melt channels to their center points. The silicate-melt network connectivity (α) was analyzed using statistical results (Fig. 2b, d and f) of the coordination number of nodes (N , the number of one node connected to neighboring nodes): an N of 5 or higher represents either physical junctions in which more than four grains are present or artifacts; these nodes with coordination numbers of 4 show an ideal interconnected melt network (von Bargen and Waff, 1986); but $N = 3$ implies locally interconnected melt channels that are more common in the nonideal system; the nodes of $N = 2$ are removed from the skeleton since two connected melt conduits effectively act as one single conduit (Miller et al., 2014); finally, $N = 1$ indicates “dead-ends” of melt channels or thin melt films that may not be resolved due to the limited resolution of microtomography. In our experiments containing multiple solid phases, nodes with a coordination number of 1 were most abundant (Fig. 2b, d and f), thereby showing a nonideal locally interconnected silicate-melt network lacking the well-established long-range connectivity. Additionally, with increasing temperature, the cumulative frequency of $N = 3$ and 4 increased from $23.1 \pm 1.4\%$ at 1200 °C ($m = 3581$) to $26.5 \pm 1.5\%$ at 1250 °C ($m = 3432$) based on large sample statistics of the node coordination number (Fig. 2d and f).

3.3. Morphology of two immiscible liquids under deformation conditions

Shear experiments (Table 2) with the starting material of olivine + basalt (10 wt%) + sulfide (5 wt%) were used to explore and determine the effect of deviatoric stress on the extraction and segregation of two immiscible liquids in a mantle peridotite. Driven by deviatoric stress, stranded sulfide droplets in static experiments were elongated and transported along with silicate melt forming some liquid-rich sheets (Fig. 3 and 4a) that interconnected a significant fraction of silicate melt and initially isolated sulfide liquid. These liquid-rich sheets were oriented at $14.3 \pm 4.5^\circ$ to the shear plane and antithetic to the shear direction with a length of several hundred microns. Notably, the angle of liquid-rich sheets did not show an apparent change with increasing strain from 1.6 (GA394) to 2.5 (GA383). These liquid-rich sheets tended to converge into low-permeability contact regions between the sample and Al₂O₃ pistons (Fig. 3a). Additionally, some sheets consisting only of elongated sulfide droplets were observed and oriented at a slightly higher angle ($29.9 \pm 3.6^\circ$) (Fig. 4b). Several small sulfide droplets (diameter $< 2.9 \pm 0.1$ μm, the aspect ratio < 2) still retained roughly spherical shapes and were distributed like beads decorating the shear zone (Fig. 4c). The EBSD mapping analyses (Fig. S5) indicated that olivine developed a diffusion fabric (CPOs) characterized by the [100] and [001] axes forming a girdle in the foliation plane (shear plane) and the [010] axes forming high concentrations approximately normal to the shear plane (Fig. S5a) (Holtzman et al., 2003). And, olivine grains also showed strong SPOs oriented 135–158° along the shear direction (Fig. S5b). In addition, a few orthopyroxene (Opx)-aggregate bands (Fig. 3a and Fig. S6) produced by the olivine-melt reaction were observed in the sample, and there were a small number of very small sulfide microdroplets (average size $\sim 0.9 \pm 0.5$ μm) (Fig. S6b) in these bands. The orientation of these bands increased from approximately 147° to 158° along the shear direction during progressive deformation as the strain increased from 1.6 to 2.5, which was apparently inconsistent with the direction of the liquid extraction and may represent the relic of silicate melt flowing (Soustelle et al., 2014).

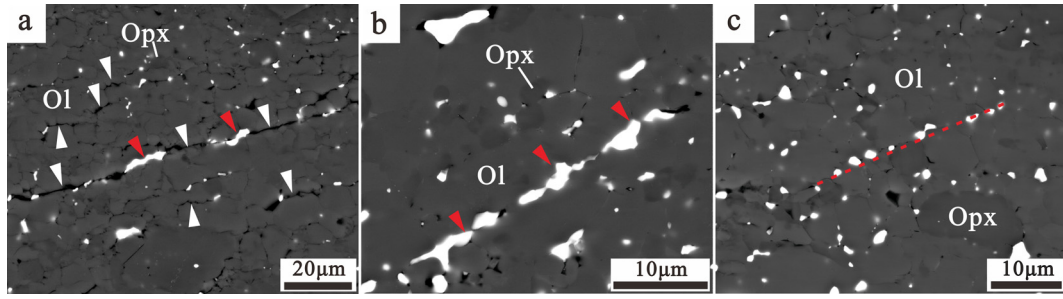


Fig. 4. Microstructures (back-scattered electron images) of experiment GA383. (a) Liquid-rich sheets consisting of elongated silicate melt (white arrows) and sulfide pockets (red arrows). (b) An only sulfide-bearing (red arrows) sheet oriented at a slightly higher angle ($\sim 30^\circ$) to the shear plane. (c) The beading distribution of small sulfide droplets (red dotted line) with no significant change in morphology. Mineral abbreviations: Ol-olivine; Opx-orthopyroxene. The shear direction is the same as Fig. 3.

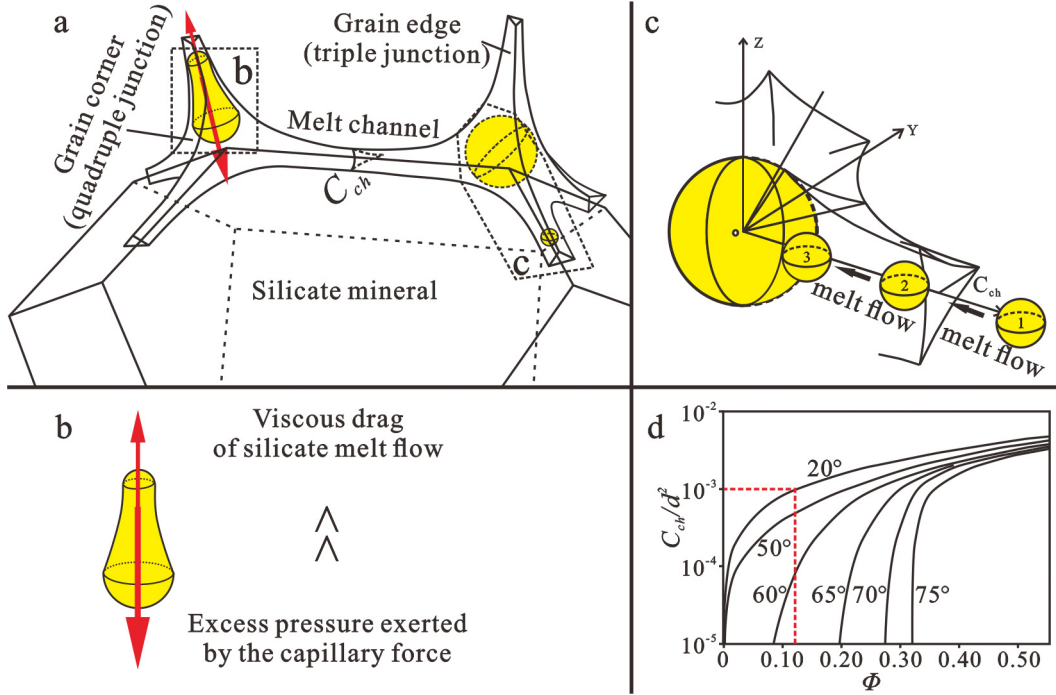


Fig. 5. Schematic diagrams (a, b, and c) of an ideal partially molten system containing sulfide droplets (yellow balls) and the theoretical calculation model (d) of normalized minimum channel cross-sectional areas (C_{ch}) (von Bargen and Waff, 1986). (a) An ideal melt network containing sulfide droplets. The polyhedron represents silicate mineral. (b) A sulfide droplet larger than C_{ch} is affected by the capillary force and the viscous drag of silicate melt flow. The excess pressure exerted by the capillary force is much larger than the viscous drag of silicate melt flow based on the analyses of Mungall and Su (2005) and the interfacial tension (~ 0.24 J/m²) between sulfide liquid and silicate melt calibrated by Mungall et al. (2015). (c) Cut-away view of a grain corner unit showing the migration process (from 1 to 3) of very small sulfide microdroplets (the size $< C_{ch}$) along with the silicate melt flow. (d) Normalized minimum channel cross-sectional areas (C_{ch}/d^2) versus melt fraction (Φ). The red dotted line marks the value of $C_{ch}/d^2 \sim 10^{-3}$ when Φ and dihedral angle (θ) are respectively 0.12 and 20° .

4. Discussion

4.1. Hydrostatic distribution and extraction of two immiscible liquids

The melt extraction from a matrix begins with the grain-scale porous flow and is largely affected by the geometry of the melt distribution (Faul, 2001; von Bargen and Waff, 1986). To quantitatively analyze the melt distribution, some theoretical parameters are cast to describe the physical properties of the melt network, such as permeability (k) and melt extraction velocity (v) (von Bargen and Waff, 1986). A simple power-law relationship between permeability (k), grain size (d) and melt fraction (Φ) has been derived for an ideal porous rock consisting of quadruple junctions connected by four narrow triple junctions (Fig. 5a) (von Bargen and Waff, 1986):

$$k = \frac{\Phi^n d^2}{C}, \quad (2)$$

where C and n are parameters that depend on the topology of the melt phase and the geometry of an individual melt channel. For a system with a uniform distribution of silicate melt, the value of n in equation (2) mainly ranges from 2 to 3 (Fig. 6) (e.g., von Bargen and Waff, 1986; Wark and Watson, 1998), and $n = 2$ is more appropriate for an isotropic system with uniform grain size and shape and a relatively low melt fraction (Connolly et al., 2009). For more complex systems with pore geometry deviating from the ideal case and a higher melt fraction, higher n values ($n = 3$) should be used (Wark and Watson, 1998; Zhu et al., 2011). Besides, for a system with an interconnected sulfide-liquid network, Roberts et al. (2007) suggested that $n = 3.6$ is more suitable under hydrostatic conditions. Moreover, the value of C in equation (2) is more controversial and recommended values range over three orders of magnitude from 3 to 3000 (Fig. 6) (e.g., Connolly et al., 2009; McKenzie, 1989; Watson and Roberts, 2011). Recently, Miller et al. (2014; 2016) obtained more exact experimental estimations for the parameter $n = 2.4$ – 2.8 (mean value ~ 2.6) and

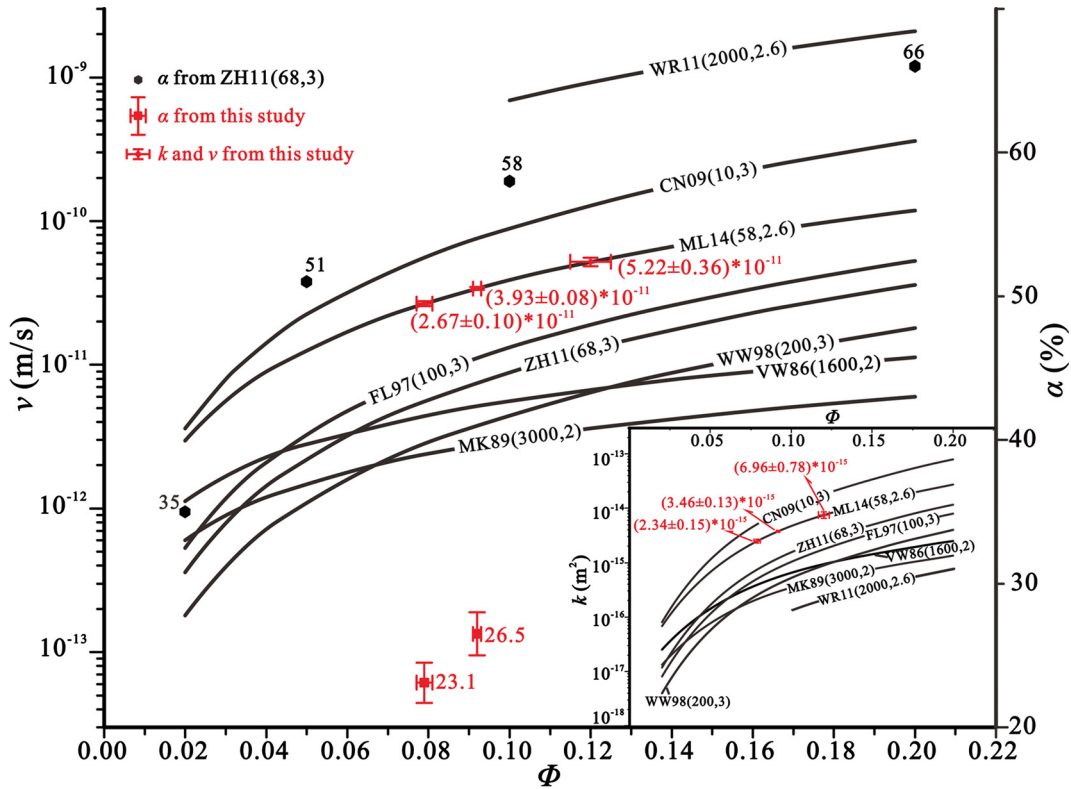


Fig. 6. Silicate-melt network connectivity (α) and extraction velocity (v) of liquids as a function of porosity (Φ) under hydrostatic conditions. The cumulative frequency of $N = 3$ and 4 in the melt network skeleton represents the α of silicate melt. The v of silicate melt is calculated with $d \sim 10 \mu\text{m}$, $\Delta\rho_m \sim 450 \text{ kg/m}^3$, $\eta_m \sim 5 \text{ Pa s}$ using various calculation models of equation (2) with different C and n values for comparison. The inset is the corresponding permeability (k). The Φ used to calculate the k and v in this study is $\sim 7.87 \pm 0.19\%$, $\sim 9.17 \pm 0.13\%$ (from three-dimensional image analyses) and $12.00 \pm 0.52\%$ (from two-dimensional image analyses). Numbers in parentheses of curve labels indicate values C and n ; for example, VW86 (1600, 2) represents $C = 1600$ and $n = 2$ from von Bagen and Waff (1986), and in the same way, MK89 (3000, 2), FL97 (100, 3), WW98 (200, 3), CN09 (10, 3), ZH11 (68, 3) and ML14 (58, 2.6) are from McKenzie (1989), Faul (1997), Wark and Watson (1998), Connolly et al. (2009), Zhu et al. (2011) and Miller et al. (2014) for silicate melt, respectively. WR11 (2000, 2.6) is from Watson and Roberts (2011) for the interconnected sulfide liquid ($\Phi > 0.10$; $\eta_s \sim 0.02 \text{ Pa}\cdot\text{s}$).

$C = 36\text{--}94$ (mean value ~ 58) by using the three-dimensional melt analysis technology to numerically simulate the Stokes flow of silicate melt and measure the permeability of olivine-basalt and olivine-orthopyroxene-basalt aggregates. In this study, both two- and three-dimensional liquid distributions reveal that silicate melt is dominated by tubular melt channels along grain edges, probably generating a more effective percolation than melt films along grain boundaries (e.g., Zhu et al., 2011). The high frequency of $N = 1$ in the topology of three-dimensional liquid distributions indicates that silicate melt forms locally interconnected networks with high tortuosity, which deviates from the ideal system. The presence of multiple solid phases each with anisotropic interfacial energies and the small grain size of silicate minerals increasing the tortuosity of melt channels may be main reasons for the high frequency of $N = 1$ and tubular melt channels with convexities in three-dimensional liquid distributions, thereby pointing to an effective mantle permeability rather lower than in the ideal case (Miller et al., 2016). With increasing temperature, the increasing frequency of $N = 3$ and 4 and the decreasing silicate melt-mineral dihedral angle (Takei and Shimizu, 2003) (Fig. 2d, f and Fig. S4) are both indicative of the increasing melt connectivity and permeability. In the present study, there are no distinct differences in liquid distributions and the silicate-melt fraction between sulfide-free (PC184) and sulfide-bearing (PC185) samples (Fig. 2a-d), thereby indicating that the presence of a small amount of sulfide liquid (1 wt%) has an insignificant effect on the topology of the silicate melt network. These results both imply that the permeability of the sulfide-bearing system in hydrostatic experiments can be described by the

theoretical model that remains a good approximation for low melt fraction systems ($\Phi \sim 0.02$ and perhaps lower) (Zhu et al., 2011). Thus, for the polyphase mineral assemblage representative of partially molten peridotite with a locally interconnected silicate-melt network in this study, the values of C (36–94) and n (2.4–2.8) suitable for a multiphase-mineral system in Miller et al. (2014) are used to calculate the permeability. For $\Phi \sim 0.07\text{--}0.12$ and $d \sim 10 \mu\text{m}$ in these hydrostatic experiments, the calculated permeability k is $10^{-14}\text{--}10^{-16} \text{ m}^2$ for silicate melt (Fig. 6), whereas in mantle rocks with a low melt fraction ($\Phi \sim 0.02$) and a large grain size ($d \sim 3 \text{ mm}$), k would be $10^{-11}\text{--}10^{-12} \text{ m}^2$, which is similar to previous estimates ($10^{-11}\text{--}10^{-12} \text{ m}^2$) (Connolly et al., 2009; Zhu et al., 2011). Based on the numerical model of McKenzie (1989), for a Darcy-type flow with the constant porosity and without compaction, the extraction velocity (v) of silicate melt with respect to a stationary matrix is estimated by:

$$v = \frac{k \Delta\rho_m g}{\Phi \eta_m}, \quad (3)$$

where $\Delta\rho_m \sim 450 \text{ kg/m}^3$ is the density contrast between silicate melt and solid upper mantle (pyrolite), $g \sim 10 \text{ m/s}^2$ is the acceleration due to gravity, and $\eta_m \sim 5 \text{ Pa}\cdot\text{s}$ is the melt viscosity (Bockrath et al., 2004). Thus, v is $0.7\text{--}11.1 \mu\text{m/day}$ ($\Phi \sim 0.07\text{--}0.12$) in hydrostatic experiments and $237.7\text{--}2967.8 \text{ cm/year}$ ($\Phi \sim 0.02$) in nature, which is within the range of the previous estimate ($200\text{--}15000 \text{ cm/year}$) (Connolly et al., 2009) but almost two orders of magnitude higher than the other calculated extraction velocity (50 cm/year) (Zhu et al., 2011) (Fig. 6). The various calculation models of k with the wide range of chosen parameters is the main

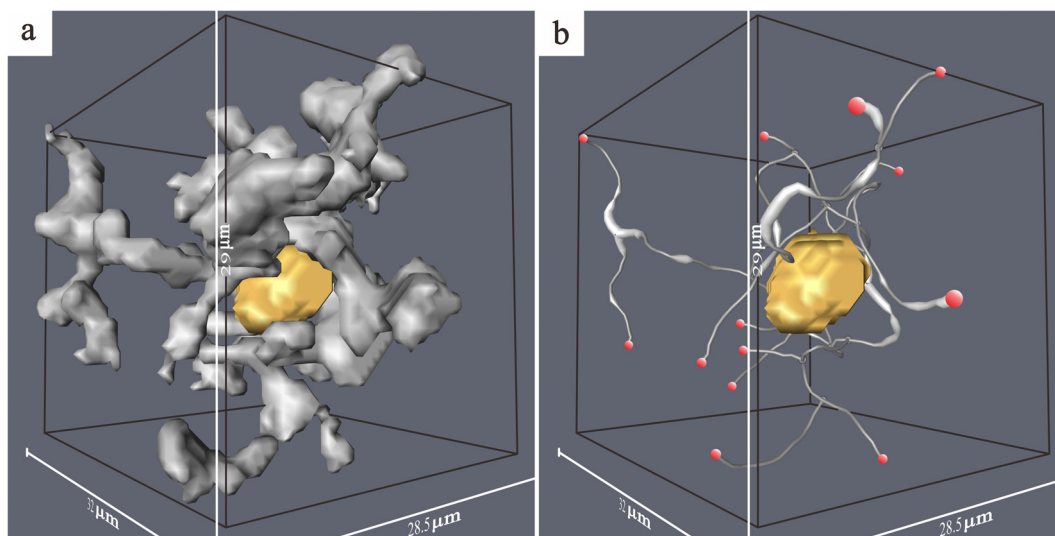


Fig. 7. A single interconnected melt-network unit (a) and its skeletonization result (b) in experiment PC226. Colors are as in Fig. 2. Red spheres in (b) represent dead ends of melt channels. The corresponding video is in supplementary files (Video S2).

reason of these contradictory values of v (Fig. 6) (e.g., Connolly et al., 2009; Zhu et al., 2011), which implies that the accurate determination of the melt distribution is vital to estimating melt velocity (Faul, 1997; Miller et al., 2014). In contrast, the connectivity (α , represented by the cumulative frequency of $N = 3$ and $4 \sim 26.5\%$ at $\Phi \sim 0.09$) of the high- v melt network in this study is considerably lower than that of the previous study ($\alpha \sim 58\%$ at $\Phi \sim 0.10$, Zhu et al., 2011) with a lower v at a similar melt fraction (Fig. 6). To reconcile the contradiction between the high v and the low α of silicate melt, this study extracted the elaborate topology of a single interconnected melt network (Fig. 7a and Video S2 in supplementary files) to exactly determinate the distribution of two immiscible liquids. A few “breakpoints” ($N = 1$; Fig. 7b-red spheres) can be observed after skeletonization for the single interconnected melt network, and most of them are potentially dead-ends of melt channels due to the presence of multiple solid phases with the small grain size (Miller et al., 2016), but the contribution of thin melt films that were not resolved due to the limited resolution cannot be completely ruled out especially under high melt-fraction conditions (see the methodology of three-dimensional image analyses in the supplementary material). In either case, the k and v of silicate melt at these positions will be negligible and cannot be estimated by these existing theoretical models, thus the values of k and v calculated by this study should only be used to characterize the single interconnected melt network, not the large-scale system representative of the whole rock. Therefore, our results suggest that the distribution and extraction of silicate melt in a partially molten peridotite are more heterogeneous and complex than the simple system and may not be simply described by the ideal model, which probably over-estimates effective permeability considerably.

For Fe-S or Fe-S-O sulfides, the wide range of possible percolation thresholds (3–17.5 vol%) is sensitive to the composition of sulfides (especially the content of oxygen in sulfide liquids) (Bagdasarov et al., 2009a; Gaetani and Grove, 1999; Terasaki et al., 2005; Yoshino et al., 2003). In hydrostatic experiments, the Fe-Ni-Cu sulfide liquid ($\theta \sim 91.5\text{--}101.3^\circ$) does not form an interconnected network, implying that the maximum content of sulfide (3.77 vol%) in this study is lower than the percolation threshold of Fe-Ni-Cu sulfide, which is consistent with predictions (4.5–5.5 vol%) of numerical models for pinch-off boundaries (e.g., von Bagen and Waff, 1986). Due to the presence of silicate melt, sulfide droplets are partially shielded by silicate melt in hydrostatic experiments,

which requires a higher fraction of sulfide liquid to overcome a higher interconnectivity threshold (Yoshino et al., 2004). Furthermore, because the viscous drag of silicate melt flow is almost three orders of magnitude lower than the excess pressure exerted in sulfide droplets by the capillary force that tends to retard its entrance into the pore throat (Fig. 5a and b) (Mungall et al., 2015; Mungall and Su, 2005), only the sulfide droplets with sufficiently small size (microdroplets) to fit through the smallest dimension of grain-edge channels, namely the minimum channel cross-sectional area (C_{ch} , the middle of melt channels in an ideal melt network) (Barnes et al., 2017; von Bagen and Waff, 1986), can be entrained along with the silicate melt flow (Fig. 5c). Providing a C_{ch} of $0.1 \mu\text{m}^2$ calculated with a dihedral angle ($\theta \sim 20^\circ$), grain size ($d \sim 10 \mu\text{m}$) and melt fraction ($\Phi \sim 0.12$, the largest value in this study) (Fig. 5d), the diameter of sulfide microdroplets entrained along with silicate melt is smaller than approximately $0.3 \mu\text{m}$ (the minimum channel cross-section and the sulfide microdroplet are assumed as an equilateral triangle and an inscribed circle, respectively), thereby indicating that under hydrostatic conditions, most sulfide droplets ($> 0.3 \mu\text{m}$) would be stranded and choked in melt junctions or pockets (Fig. 5a). Moreover, it is conceivable that under longer timescale conditions, large sulfide droplets tend to grow through diffusion-controlled Ostwald ripening until they are large enough to block the pore throat of melt channel, thereby further prohibiting the migration of sulfide droplets under static conditions (Mungall and Su, 2005) as observed distributions of sulfide pockets in mantle peridotites (e.g., Lorand et al., 2010).

In conclusion, due to these restrictions of the melt-channel smallest dimension and the prevalence of dead ends in the locally interconnected melt network, the extraction and segregation of sulfide liquid are inefficient and largely inhibited by silicate melt in a hydrostatic partially molten peridotite (Fig. 5a and 8).

4.2. Dynamic extraction of two immiscible liquids

Based on natural observations and previous shear-deformation experiments, the transport of silicate melt in some regions of the partially molten mantle is known to feature strong channelization caused by porous-media flow instability, thereby developing high-permeability pathways for the extraction of silicate melt (Holtzman and Kohlstedt, 2007; Kelemen et al., 1995). Stress-driven melt segregation (Holtzman and Kohlstedt, 2007) is one of the main driving forces for flow instability and can cause an order-of-magnitude increase in the melt extraction velocity (Connolly et al., 2009). Driven

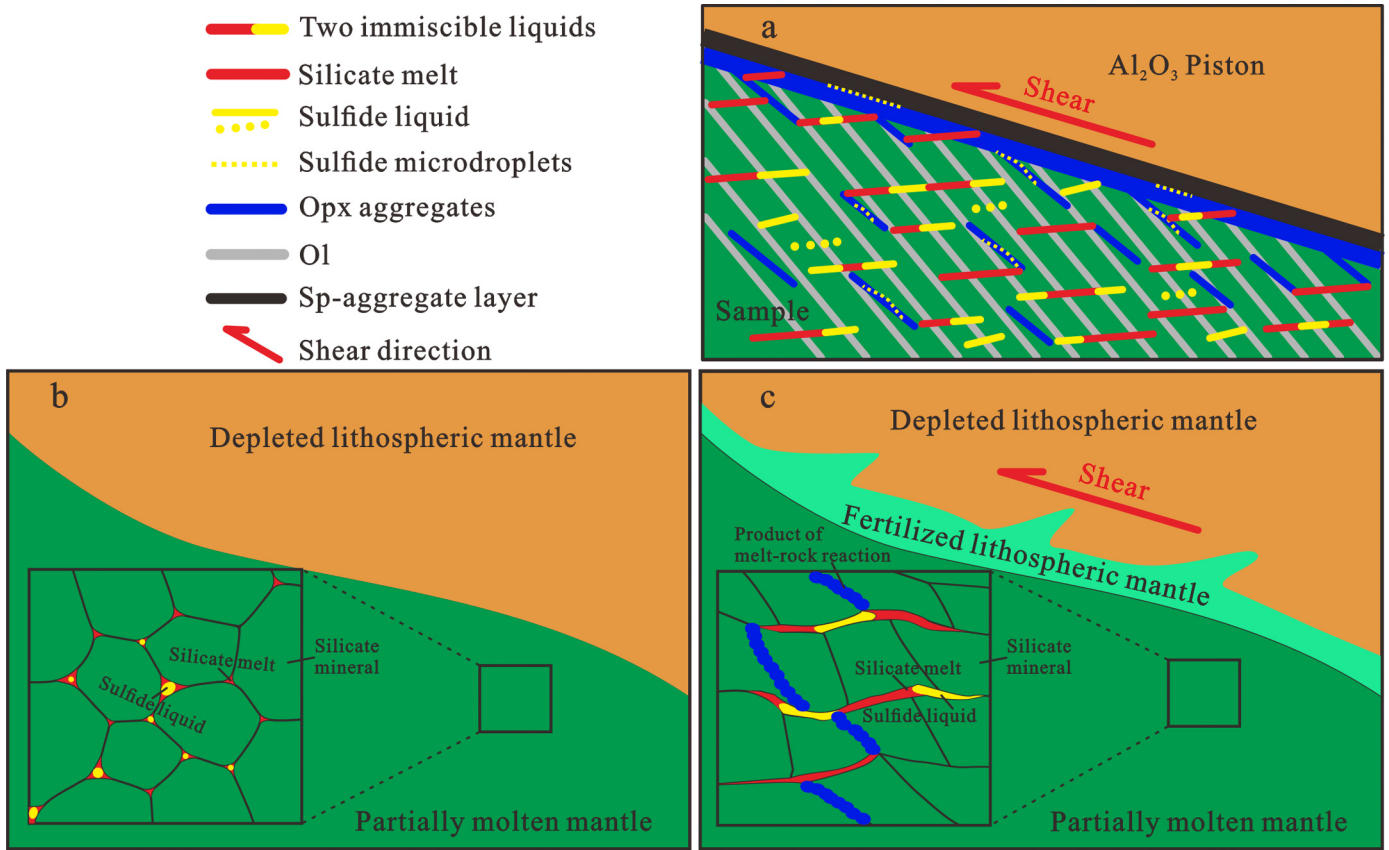


Fig. 8. Cartoon maps illustrating the distribution and extraction of two immiscible liquids under hydrostatic and dynamic conditions. (a) The distribution of two immiscible liquids and the shape preferred orientations (SPOs) of silicate minerals in shear experiments. Silicate melt can react with the Al_2O_3 piston to produce a spinel-aggregate layer (black bold line). (b) and (c) Illustrations of corresponding expected natural counterparts displaying that initially stranded sulfide droplets in the hydrostatic mantle (b) are elongated and extracted along with silicate melt to fertilize the lithospheric mantle during ductile deformation of the partially molten mantle (c). The content of sulfide in Fig. 8b and c is exaggerated to highlight the distribution and extraction of sulfide liquid. Mineral abbreviations: Ol-olivine; Opx-orthopyroxene; Sp-spinel.

by shear stress, sulfide liquids with high percolation thresholds can also be efficiently extracted and even drained from solid silicates along liquid-rich sheets or aligned elongated liquid pockets to obtain a low “pinch-off” liquid fraction $\sim$ two orders of magnitude lower than the percolation threshold (Groebner and Kohlstedt, 2006; Hustoft and Kohlstedt, 2006). In contrast, under low-strain conditions (strain ~ 0.15 – 0.35 , strain rate $\sim 10^{-3}$ – 10^{-5} s^{-1}), previous studies (Cerantola et al., 2015; Walte et al., 2011) suggested that the presence of silicate melt hinders the extraction of sulfide liquid by reducing its connectivity, which is consistent with the higher percolation threshold of sulfide liquid under static silicate melt-bearing conditions as we document above (see also Yoshino et al., 2004), so that the stress-driven extraction of sulfide liquid at low sulfide-liquid fractions (5 wt% or less) can occur only under silicate melt-free conditions. Comparing with those hydrostatic and low-strain deformation experiments, in our high-strain shear experiments (strain ~ 1.6 – 2.5 , strain rate $\sim 10^{-4}$ s^{-1}) containing two immiscible liquids, the angle ($14.3 \pm 4.5^\circ$) of liquid-rich sheets is similar to the results (15 – 20°) of systems containing only silicate melt (e.g., Holtzman and Kohlstedt, 2007) but slightly lower than that (18 – 30°) of systems containing only sulfide liquid (e.g., Hustoft and Kohlstedt, 2006), indicating that silicate melt dominates the liquid alignment. That is, in the presence of silicate melt, the shearing response of the entire aggregate to sulfide liquid is similar to that of partially molten silicate aggregates, which is likely because the dynamic viscosity of sulfide liquids ($\eta_s \sim 0.02$ Pa s, calculated using the model of Terasaki et al., 2001) is several orders of magnitude lower than that of silicate melts. The lower angle is attributable to the positive feedback between the anisotropic

viscosity (stress-dependent viscosity) leading to liquid segregation into the silicate melt-dominated liquid alignment and the liquid-preferred orientation enhancing the viscous anisotropy (Takei and Katz, 2015). The open channels develop where the applied shear stress acts locally at the tip of the triple junction. The localized tensile stress concentration is relaxed when silicate melt extends into and wets the appropriately oriented grain boundary between silicate minerals (Kohlstedt and Holtzman, 2009). The opened grain boundary by silicate melt provides a pathway with a higher porosity (Φ , the total porosity of silicate melt and sulfide liquid in liquid-rich sheets) for the efficient extraction of sulfide liquid with the low dynamic viscosity (Fig. 4a and Fig. 8a).

During the extraction of two immiscible liquids, solid phases also undergo corresponding changes in their crystallographic and morphological orientation, such as the fabric characteristics (CPOs and SPOs) of the olivine and Opx-aggregate bands (Fig. 8a, Fig. S5 and Fig. S6). Oriented liquid-rich sheets provide fast-diffusion pathways for solid components (Holtzman et al., 2003), thereby enhancing diffusion-accommodated creep and resulting in the diffuse CPO pattern of olivine (Fig. S5a). This finding implies that the contribution of diffusion-accommodated creep to mantle flow may be higher than previous thoughts due to the presence of liquid-rich sheets (Holtzman et al., 2003). Moreover, the consistent association of sulfide microdroplets (0.9 ± 0.5 μm) with Opx-aggregate bands indicates that driven by stress, larger sulfide microdroplets than those under hydrostatic conditions (0.3 μm , see section 4.1) are entrained along with silicate melt flow (Fig. 8a-yellow dotted lines and Fig. S6).

When stress causes the alignment of silicate melt and sulfide pockets into sheets, the permeability becomes highly anisotropic being highest in the direction parallel to and lowest in the direction normal to the alignment. As two immiscible liquids reside in a set of parallel sheets rather than tubes, we adopt values of $n = 3$ and $C = 12$ or 25 in equation (2) (Holtzman and Kohlstedt, 2007; Hustoft and Kohlstedt, 2006) to calculate the permeability (10^{-15} m^2) for liquid flowing parallel to liquid-rich sheets ($\Phi \sim 0.10$, $d \sim 5 \text{ }\mu\text{m}$). Based on two-phase flow theory (Mckenzie, 1984), the calculation model of velocity in equation (3) is usually applied to the state of uniform fluidization. During shear deformation, the extraction velocity of a melt is mainly controlled by the matrix deformation rate and the matrix viscosity and may be still controlled indirectly by the permeability and the melt viscosity as expressed by equation (3). Matrix viscosity is highly anisotropic and varies dynamically with shear stress (Takei and Katz, 2015), thus the extraction velocity is also highly anisotropic and cannot be estimated by the simple theoretical model. The mobility (inversely proportional to the viscosity) of low-viscosity sulfide liquid is higher than that of the silicate matrix. During liquid extraction, the pressure gradient caused by the antithetic material flow of the silicate matrix (the SPOs of olivine and opx-aggregate bands) (Fig. 8a) may further drive the extraction of the high-mobility sulfide liquid from the compression side to the tension side along with these high-permeability liquid-rich sheets (Kohlstedt and Holtzman, 2009). Therefore, using the calculation model of equation (3) without considering the effect of stress (Hustoft and Kohlstedt, 2006), this study estimates a possible lower limit of the extraction velocity (475.2–990.0 $\mu\text{m/day}$) for sulfide liquid along these liquid-rich sheets calculated with $\Delta\rho_s \sim 1100 \text{ kg/m}^3$ (Table 1, Kress et al., 2008) and $\eta_s \sim 0.02 \text{ Pa}\cdot\text{s}$ (Terasaki et al., 2001) in shear experiments. In nature, stress can potentially drive the focused porous flow of melt through high- Φ channels (Holtzman and Kohlstedt, 2007) that concomitantly leads to a much faster v for the transport of sulfide liquid based on the relationship of $v \propto \Phi^{n-1}$ derived from equation (3), but the exact v of sulfide liquid may not be estimated due to the unknown permeability structure of melt-rich channels.

In conclusion, during shear deformation to large strains, silicate melt-dominated liquid-rich sheets can create a high-velocity pathway potentially for not only the diffusion of solid components but also the efficient extraction and segregation of initially stranded sulfide droplets along with the flow of silicate melt through a deformable partially molten peridotite.

5. Implications for the fertilization of lithospheric mantle and the genesis of magmatic sulfide deposits

The genesis of magmatic Ni-Cu sulfide deposits is closely associated with the structure and composition of the lithospheric mantle under cratons (Arndt et al., 2005). It has been suggested that the lithospheric mantle is commonly fertilized by the reaction-infiltration of sulfide-bearing melts, which may contribute to the ore-metal budget of magmatic sulfide deposits (Griffin et al., 2004), such as Ni-Cu-PGE sulfide deposits of the Bushveld Complex (the largest magmatic PGE deposits in the world) in the Kaapvaal craton (Griffin et al., 2004) and the Jinchuan Ni-Cu sulfide deposit (the largest single magmatic Ni-Cu deposit in the world) in the North China Craton (Li et al., 2005).

Whereas most studies of the removal of sulfide liquid from a mantle peridotite undergoing partial melting have focused on the need for sulfide liquid to be progressively dissolved by the departing silicate melt (e.g., Holzheid and Grove, 2002; Mungall and Brennan, 2014), the present study suggests a feasible process by which sulfide liquid can be at least partially removed mechanically while the restite assemblage remains saturated with the sulfide liquid

phase. This process could more easily explain the refertilization of the depleted lithospheric mantle, by reintroducing ore-forming elements that eventually might be tapped to form magmatic sulfide deposits. As silicate melt is extracted into liquid-rich sheets, the deviatoric stress would be directly imposed on sulfide liquid because most of sulfide droplets are not surrounded and shielded by silicate melt. Thus, scaling from this study (strain rate $\dot{\epsilon} \sim 10^{-4} \text{ s}^{-1}$; deviatoric stress $\sigma \sim 137\text{--}161 \text{ MPa}$; sulfide droplet radius of curvature $r \sim \mu\text{m}$ -range) to the natural mantle, we need to extrapolate the effect of deviatoric stress on sulfide droplets to natural mantle conditions, irrespective of long-range forces such as the gravity due to the negligible magnitude comparing to the stress. When the behavior of sulfide liquid in a deformable partially molten peridotite is dominated mainly by stress rather than by surface tension, sulfide-liquid pockets will become elongated and extracted into grain boundaries opened by silicate melt. Considering the effect of stress on isolated sulfide droplets in the mantle from which silicate melt is actively being extracted, the following inequality expresses the idea that the deviatoric stress acting on a liquid pocket must exceed the effect of interfacial tension if the remaining sulfide liquid is also to be mobilized (Groebner and Kohlstedt, 2006; Kohlstedt and Holtzman, 2009), that is:

$$\frac{\sigma}{\gamma_{o-s}/r} > i, \quad (4)$$

where the parameter i equals to 1 according to Groebner and Kohlstedt (2006), γ_{o-s}/r is the surface tension expressed as the ratio of the solid (olivine)-sulfide liquid interfacial energy ($\gamma_{o-s} \sim 0.6\text{--}0.7 \text{ J/m}^2$; Groebner and Kohlstedt, 2006) and the radius of curvature of sulfide droplets. The use of the olivine-sulfide interfacial energy and the neglect of the sulfide liquid-silicate melt interfacial energy in this expression is justified by the assumption that relatively large sulfide droplets in pockets being drained of silicate melt will end up being pressed against the solids over most of their surface areas. In our deformation experiments, shape changes of small roundish sulfide droplets (diameter $< 2.9 \pm 0.1 \text{ }\mu\text{m}$) are still resisted by the surface tension. Under a given stress condition, a proportionality relation between r and $\dot{\epsilon}$ can be derived from inequality (4):

$$r \propto \dot{\epsilon}^{-\frac{1}{w}}, \quad (5)$$

where w is stress exponent (Walte et al., 2011). Extrapolating to natural mantle conditions (lower $\dot{\epsilon} \sim 10^{-12}\text{--}10^{-14} \text{ s}^{-1}$; lower $\sigma \sim 0.1\text{--}1 \text{ MPa}$; larger $r \sim \text{mm-to-cm}$ range) based on the (4) and (5), sulfide droplets with the size $> \text{mm-scale}$ can be elongated and extracted by the stress. Considering the low content of sulfide ($< 0.1 \text{ wt\%}$) in fertile mantle peridotites (Lorand et al., 2010), the mixture density of two immiscible liquids in a liquid-rich sheet is commonly lower than the surrounding mantle when the fraction of sulfide liquid in the liquid-rich sheet is lower than about 26.5 wt% based on the calculation of mass balance. Therefore, driven by the shear stress, large ($> \text{mm-scale}$) sulfide droplets could be extracted along silicate melt-dominated liquid-rich sheets from the deformable partially molten mantle into the lithosphere to fertilize the lithospheric mantle and cause the primary enrichment of sulfides for magmatic sulfide deposits, which may be the first step of the sulfide ore formation (Fig. 8b and c).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We are grateful to two anonymous reviewers for their useful suggestions and comments, and to Prof. Meifu Zhou and Prof. Yao Wu for discussions. We thank Prof. Yanbin Wang and Dr. Feng Shi for help with X-ray synchrotron microtomography analysis, Dr. Wenlong Liu for help with EBSD analysis. This research was supported by the National Natural Science Foundation of China (No. 40172068). This research used resources of the Advanced Photon Source, a U.S. Department of Energy (DOE) Office of Science User Facility operated for the DOE Office of Science by Argonne National Laboratory under Contract No. DE-AC02-06CH11357.

Appendix A. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.epsl.2020.116162>.

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